IA-CDNet: Change Detection in Adverse Remote Sensing Image Conditions With an Advanced Image-Adaptive Method

Zhipan Wang, Zijun Yang, and Qingling Zhang

***Abstract*— Change detection using deep learning methods in optical remote sensing imageries has become the mainstream method, primarily due to their strong spatial–temporal transferability. However, current change detection models are designed for high-quality optical remote sensing imageries and seldom account for image conditions such as haze cover or low-light scenarios. As we know, the quality of optical remote sensing imageries can be severely degraded by haze or low-light conditions, which in turn significantly impairs the performance of change detection models. Yet there is still a lack of advanced change detection models designed for the aforementioned adverse scenarios at present. To address this gap, we proposed a simple image-adaptive change detection model named IA-CDNet, aimed at improving model performance in adverse image conditions. Initially, we developed an image-adaptive filter module, which employed several differentiable image filter operators to enhance the quality of the inputted image. Notably, the parameters of these filters are automatically learned using a lightweight convolution neural network (CNN) module. Then, a generic change detection model is integrated to extract low-level and high-level semantic information, facilitating the detection of changes between bi-temporal remote sensing imagery. Ultimately, experiments conducted on three open-sourced change detection datasets and two large-scale real-world scenarios have demonstrated the effectiveness of IA-CDNet. We anticipate this model can make an important contribution to change detection research fields and improve model performance further. The Open sourced code can be accessed at: https://github.com/wzp8023391/Change-detection-in-adverse-conditions.**

***Index Terms*— Adverse image conditions, change detection, deep learning, image-adaptive, remote sensing imagery.**

I. Introduction

USING remote sensing images to detect landcover changes plays an important role in landcover mapping [1], urbanization research [2], [3], global climate change [4], among others. Traditional change detection tasks on bi-temporal remote sensing images have heavily relied on visual interpretation, which is not only time-consuming but also inevitably leads to omission or commission errors [5]. Consequently, developing automatic change detection methods has garnered much attention over the years. Currently, using deep learning models to detect landcover change become the mainstream approach [6], [7], [8], [9], primarily due to their superior spatial–temporal transferability compared to traditional machine learning methods. However, the majority of recent research was interested in designing robust models for change detection tasks on high-quality image conditions [10], [11], with less emphasis on enhancing model performance under adverse image conditions, such as the haze-cover or low-light conditions (an example of adverse remote sensing image conditions is depicted in Fig. 1). As we know, the quality of optical remote sensing images is inevitably degraded by the aforementioned scenarios, and the performance of deep learning models in such scenarios is significantly diminished [12].

To enhance the model performance in adverse image conditions, several methods have been proposed [12]. These approaches can be categorized into three distinct types (see Fig. 1). The first type is change detection models trained directly on original image samples [6], [13], which was the most popular used method in the current domain of change detection research. The second type uses the image enhancement method to ameliorate the image quality before applying change detection models to the preprocessed images [14], [15]. For instance, the DehazeFormer [14] and MSBDN [16] are capable of removing haze in adverse image conditions, thereby restoring high-quality images. However, these preprocessing methods will significantly prolong the model inference time, and the parameter setting is also complicated for users.

The last type involves the conjoint application of the image enhancement method and deep learning model. This kind of method has already been successfully applied to object detection under adverse image conditions. For instance, the IA-YOLO was an effective model designed for adverse image conditions [17]. In IA-YOLO, which used a lightweight convolution neural network (CNN) module to learn image filter parameters automatically to enhance night-time image quality, subsequent object detection experiments conducted on night-time images proved its effectiveness. In addition, researchers also jointly used an image enhancement module and segmentation models to improve the segmentation

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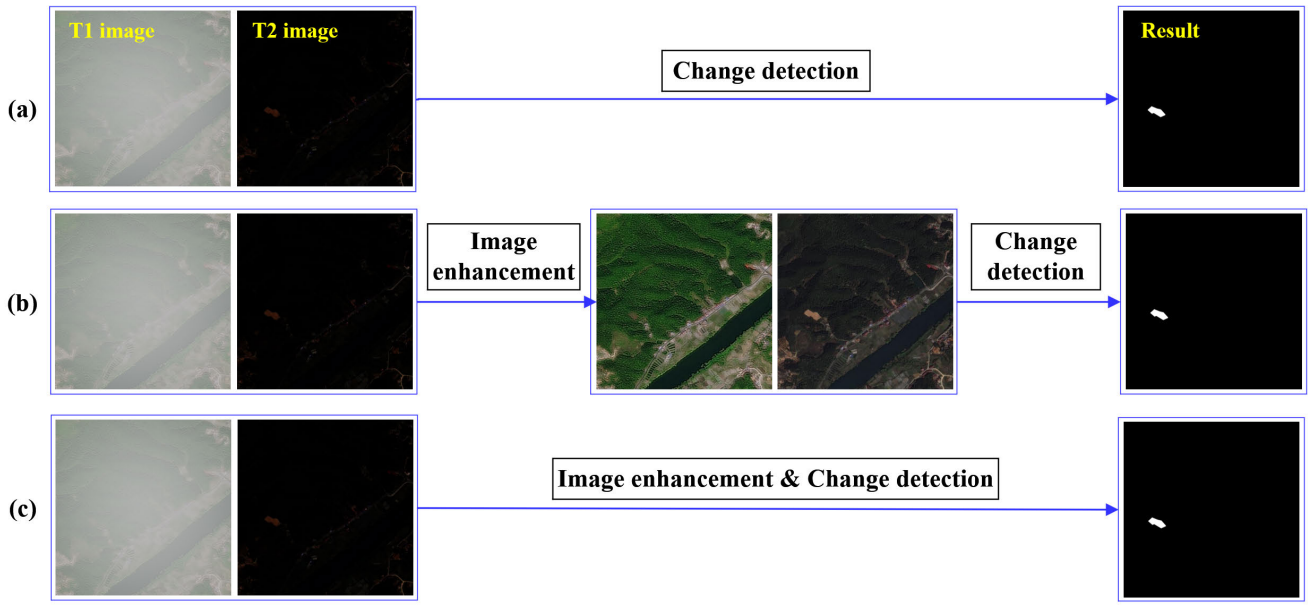


Fig. 1. Change detection in adverse image conditions. (a) Change detection models be applied directly on haze-cover or low-light image conditions. (b) Using an image enhancement method as the preprocess means and then change detection models is applied to the preprocessed images. (c) Jointly use image enhancement and change detection models in adverse image conditions.

precision for nighttime images. For example, a lightweight CNN module was used to enhance the quality of inputted images and the final segmentation results also proved its effectiveness [18]. In essence, image enhancement facilitated by self-learning CNN can be characterized as an image-adaptive method [17]. It is evident that jointly using the image enhancement method and deep learning models is simple and effective for users.

Domain adaption method can also be defined as an alternative image enhancement solution aimed at improving model performance under adverse image conditions [9], [19], [20]. For instance, the DA-MSCDNet was an excellent domain adaption method, which can be adapted to detect changes in both SAR and optical remote sensing images [19]. The SDACD introduced an interesting domain adaption method to alleviate the domain shift degeneration between bi-temporal images [20]. Most of the domain adaption methods leverage the generative adversarial network (GAN) to generate a new synthesized image based on the source image. For example, the CycleGAN was the widely used GAN for this purpose [21]. Although the domain adaption methods performed efficiently in alleviating the domain shift issue, they may still struggle in significant semantic gap scenarios [17].

Compared to domain adaption methods, the image-adaptive method possesses two distinct advantages. On the one hand, the image-adaptive method is straightforward and can readily integrated into object detection or semantic segmentation models, such as the YOLO series, DeepLab series [22], PSPNet [23], among others. On the other hand, this kind of method is its explanation and it can be defined as a useful tool to improve the image quality by using self-learning. However, despite the image-adaptive method has been successfully applied in object detection or semantic segmentation tasks for natural images successfully, there is still a lack of an advanced image-adaptive method designed for change detection tasks in remote sensing imageries.

The structure of the deep learning model is also highly associated with change detection precision. The model structures of the recent change detection models can be categorized into three categories: the CNN-based, the transformer-based, and the mixed CNN and transformer structure. The CNN-based change detection models primarily used the convolution kernel as the fundamental operator to extract low-level or high-level semantic features in bi-temporal remote sensing images. For example, the SNUnet-CD [24], SiamHRnet-OCR [25], DeepLabV3+ [26], SegFormer [27], SegNext [28], and ScribbleCDNet [29], are advanced CNN models. The transformer-based models used the basic transformer module to extract semantic features. Since the transformer can extract semantic features on the global scale of images, thus most transformer-based models demonstrated superior performance than CNN-based, such as the TCIANet [30], the AMTNet [31], and the D-LIEG [32]. However, the computation resource cost of transformer-based models is much higher than CNN-based models. Recognizing the complementary strength of both CNN-based and transformer-based change detection models, researchers also proposed hybrid CNN and transformer-style architecture. Such as the ACAHNet [33], the ConvTransNet [7], and the SMNet [10]. All of the aforementioned change detection models have made a significant contribution to the field of change detection research.

Motivated by the image-adaptive approach and sophisticated change detection model structure, we proposed a novel image-adaptive change detection model tailored for adverse image conditions in remote sensing images, named IA-CDNet. There are two important parts in IA-CDNet, including the image-adaptive filter module and the change detection module. On the one hand, we integrated a lightweight CNN module to learn image filter parameters automatically, including the haze-remove filter, exposure filter, gamma filter, contrast filter, and sharpen filter, and these image filter operators can

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enhance the image quality of bi-temporal remote sensing images automatically. On the other hand, we selected the SNUNet-CD as our foundational module for constructing the change detection module in IA-CDNet, and it is worth noting that we aimed to expand the ability of change detection models in adverse image conditions, thus the simple and effective SNUNet-CD is a suitable choice for this purpose.

The organization of this manuscript is as follows. In Section II, the detailed data and method will be described. The experiment results will be displayed in Section III. In Section IV, we will deeply discuss the proposed model. The conclusion will be presented in Section V. The main contributions of this manuscript are as follows.

1) We designed an image-adaptive deep learning model for change detection tasks in adverse remote sensing image conditions, named IA-CDNet. This model can be regarded as an efficacious approach for change detection models in adverse image conditions.
2) Three open-sourced change detection datasets and two large-scale real-world scenarios with haze-cover and low-light image conditions were selected to substantiate the performance of IA-CDNet, and the final experimental results demonstrated that IA-CDNet significantly improved the change detection accuracy.

II. Related Work

This manuscript intersects with three pivotal research domains including bi-temporal remote sensing image change detection, domain adaption, and image adaption.

A. Bi-Temporal Remote Sensing Image Change Detection

Change detection on bi-temporal images can be categorized into two types: traditional methods and deep learning methods. Traditional methods mainly use machine learning classifiers, such as the support vector machine (SVM) [34], random forest (RF) [35], maximum likelihood classifier (MLC), or the AdaBoost as the fundamental classifier to conduct change detection. These machine-learning classifiers performed well in local regions or small-scale regions. Nevertheless, their performance in complicated regions was limited, especially for spatial–temporal transferability. Deep learning methods have attracted much attention in recent years mainly because of their strong spatial–temporal transferability. Compared to traditional change detection methods, deep learning methods can learn general knowledge over a large number of training samples, which makes them perform excellent in large-scale and complicated regions.

Recently, many advanced change detection models have been proposed. For example, the SiamHRnet-OCR [25] was an excellent Siamese change detection model designed for monitoring deforestation on high-resolution remote sensing images. The SNUNet-CD change detection model acquired state-of-the-art results on an open-sourced change detection dataset [24]. The main contribution of the SNUNet-CD model was that a new ensemble channel attention module (ECAM) was proposed, which can significantly improve the model's performance. Moreover, other change detection models, such as ScanNet [36], SMDNet [37], SpectralGPT [38], SAM-CD [8], MSDFFN [39], DMINet [40], and FCD-GAN [41] also make a significant contribution for change detection research fields.

However, most of the existing research was interested in designing an advanced model structure but paid less attention to improving the model performance in adverse image conditions, such as the cloud or haze cover scenarios, cloud shadow, or mountain shadow cover scenarios. The aforementioned scenarios will significantly degrade the performance of change detection models. Therefore, developing a simple and effective method to improve change detection model performance in adverse image conditions has an important scientific and application value.

B. Domain Adaption

The domain adaption methods are widely applied in semantic segmentation, object detection, and change detection fields. The core idea of domain adaption is that decrease the domain shift degeneration between the source domain and the target domain. For instance, the CycleGAN was the most popular model designed for image-to-image transformation, and it can easily transform the style of source images into the target image to improve model performance [21]. In addition, researchers also used the domain adaption method to accomplish various style transformations, such as cross-city to cityscapes to improve segmentation precision.

Recently, domain adaption methods have also been applied in bi-temporal remote sensing image change detection. For example, the DA-MSCDNet [19] and SDACD [18] are excellent domain adaption change detection models. The DA-MSCDNet can conduct change detection on both SAR and optical remote sensing images using domain adaption. The SDACD used the image and feature domain adaption to decrease the domain shift degeneration between bi-temporal images. Despite the domain adaption method well alleviates domain shift degeneration issues, but the structure of these models is complicated, which will significantly prolong the inference time.

C. Image Adaption

The image-adaptive methods were widely applied in the image enhancement research field. Generally speaking, most of the traditional image enhancement methods need to adjust the parameter manually. Therefore, to decrease the subjective influence, several automatic methods have appeared. For example, the adaptive tune enhancement method can adjust the image brightness according to the illumination distribution of the inputted images [42]. In addition, researchers used a CNN module to learn the exposure parameter with the adversarial learning method. Meanwhile, the adaptive-edge learning module adopted deep learning to learn the mean gradient of the entire image, this model also displayed a robust performance in various scenarios [43]. Although this method can significantly improve image quality, the parameter-learning module was hard to train. Recently, a few

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TABLE I
DETAILED INFORMATION ON EXPERIMENTAL DATASETS

Datasets	Detailed information
Deforestation detection dataset [25]	This dataset was collected from the Yangtze River region of China, and the whole collected area is over 2 million square kilometers. The number of training sample images in this dataset was 8330, and the image size of each sample image is 512×512 pixels. It's worth noting that the image band of sample images only contains visible bands, and they are the red, green, and blue bands. The spatial resolution of this dataset is 2-meter. To train the deep learning models, we have split this dataset into three sub-datasets, including the training dataset, validation dataset, and test dataset, and the ratio between them is 6:2:2.
Cropland change detection dataset [11]	This dataset was provided by Wuhan University, and the number of training sample images is 3194. The image size of each training sample image is 512×512 pixels. Also, the image band of this dataset only contains visible bands. The spatial resolution of this dataset is 1-meter. We also split this dataset into three sub-datasets, including training, validation, and test datasets, and the ratio between them is the same as the deforestation detection dataset.
LIM-CD [44]	This dataset was generated for detecting newly constructed land in bi-temporal remote sensing images. The total number of this dataset is 9259, and the image size of each image pair is 512×512 pixels. In addition, the image bands of this dataset only contained visible bands. There are two label values in the label image including the primary annotation and secondary annotation. The primary and secondary annotation is the newly constructed land change between two time-phase images and constructed land in the former time-phase image, respectively. In this paper, we only want to detect changes in bi-temporal images, therefore, we only kept the primary annotation in the label image. Finally, the training, validation, and test datasets were set the same as the aforementioned two datasets.

simple image-adaptive methods have been proposed. For example, for the night-time object detection task, the IA-YOLO proposed a CNN module to learn image enhancement filter parameters automatically to improve the inputted image quality, and the final results indicated that this method can significantly improve YOLO performance in night-time conditions [17]. In addition, the researchers also proposed an image-adaption module for semantic segmentation in night-time image conditions [18], and the core idea of it is also the self-learned image enhancement filters.

The main advantage of image-adaptive methods is that they can easily be integrated into semantic segmentation or object detection models. Another advantage of this method is that it can be designed lightweight to be suitable for various tasks to improve computation efficiency, such as for automatical driving scenarios. Therefore, regarding the change detection task in adverse image conditions, the image-adaptive method may provide an effective way to improve change detection model performance in these scenarios.

III. DATA AND METHOD

A. Data

To validate the effectiveness of the proposed method, we selected three open-sourced datasets as the experimental data, and they are the deforestation detection dataset, the cropland change detection dataset, and the LIM-CD dataset. The detailed information regarding the aforementioned datasets can be seen in Table I. To validate the model performance in real-world scenarios, we also selected several bi-temporal high-resolution remote sensing images as the experimental data, and the detailed information of them can seen in Table II.

All the experimental images of real-world scenarios were acquired from the China Center of Resources Satellite Data and Application (https://www.cresda.com/zgzywxyyzx/index.html). We have used ENVI5.4 and ArcGIS10.8 to implement general image preprocessing operations, such as pansharpening, multiple image mosaic, and region clip. It is worth noting that the former time-phase images were covered with haze, cloud, or cloud shadow, which is extremely hard for visual interpretation in such adverse image conditions. Thus, we selected the reference images as the ground truth (GT) images for the former time-phase images due to the very short time intervals between reference images and the former time-phase images.

B. Methods

The flowchart of the proposed image-adaptive change detection method (IA-CDNet) can be seen in Fig. 2. The IA-CDNet can be categorized into two submodules: the image-adaptive filter module and the change detection module. The image-adaptive filter module is composed of several simple convolution modules, which were designed to learn image filter parameters. Regarding the adverse conditions in optical high-resolution remote sensing images, we used five differentiable image filter algorithms to enhance the inputted images, and they are the haze remove filter, the exposure filter, the gamma filter, the contrast filter, and the sharpen filter (the equation of these filter operators can be seen in Table III). In each original filter operator, only one parameter needs carefully manual setting. For example, for the haze remove filter operator, there is only a "W" parameter that needs to be adjusted, which can control the degree of haze removal. However, for nonexpert users, it may be hard to adjust. In IA-CDNet, we used a CNN module to learn the "W" parameter automatically, in other words, we can make the parameter adjustment process to be end-to-end. In addition, the other four parameters for the exposure filter, the gamma filter, the contrast filter, and the sharpen filter are the "E," "G," "alpha," and "numta," respectively.

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TABLE II
DETAILED IMAGE INFORMATION IN REAL-WORLD HAZE-COVER SCENARIOS

Task name	Iamge name	Data source	Image ID	Spatial Resolution	Acquired time
Deforestation detection	T1 image	GaoFen-1	20230415_L1A1228281324	2-meter	2023-4-15
	T2 image	GaoFen-2	20231017_L1A13396461001	1-meter	2023-10-17
	Reference image	ZiYuan-1E	20230416_L1B0000592506	2-meter	2023-4-16
Cropland change detection	T1 image	GaoFen-1	20190402_L1A0003918641 20190402_L1A0003918633 20190402_L1A0003918549 20190402_L1A0003918540	1-meter	2019-4-2
	T2 image	GaoFen-1C	20230406_L1A1022117633	2-meter	2023-4-6
	Reference image	GaoFen-1B	20190427_L1A1227624711	2-meter	2019-4-27

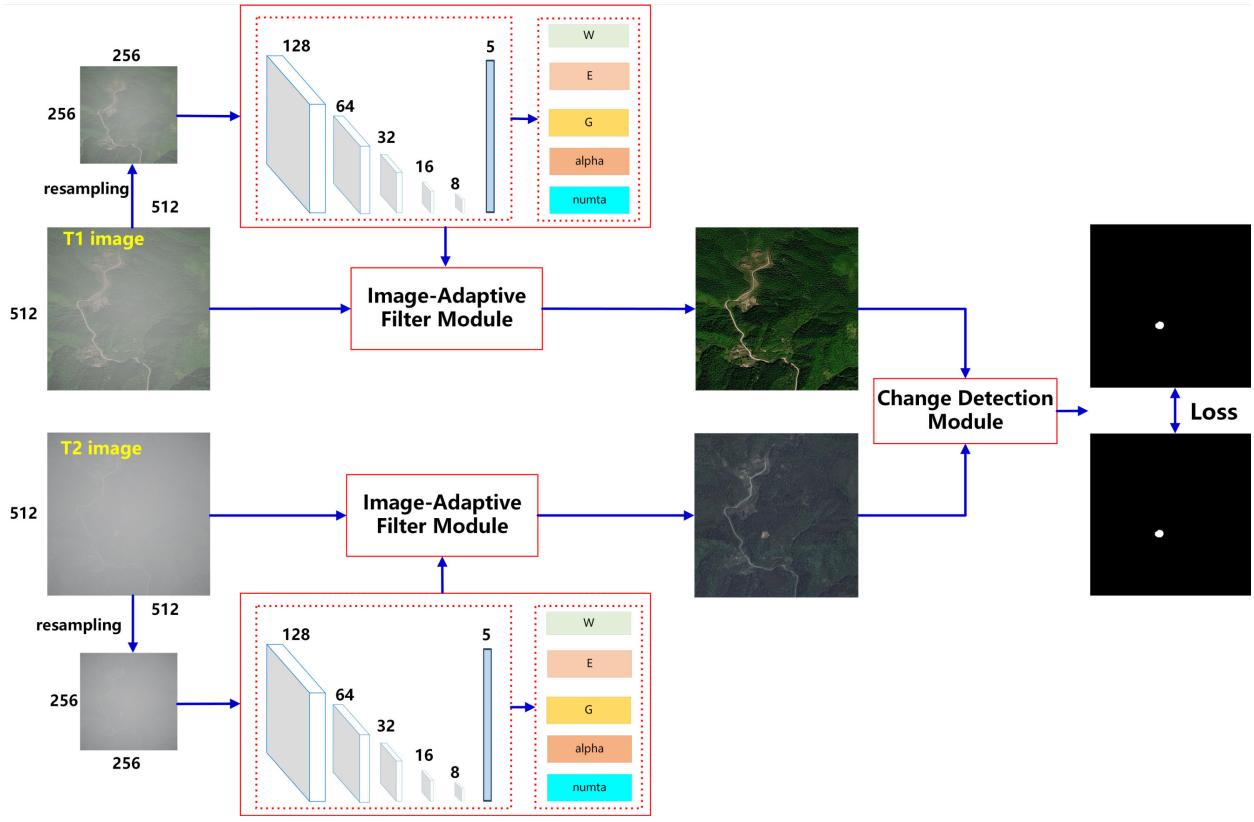


Fig. 2. Pipeline of the proposed IA-CDNet.

The change detection module in IA-CDNet can be selected from the semantic segmentation model or change detection model. In other words, the change detection module can be replaced with any other change detection model. We chose the SNUNet-CD as the change detection module in IA-CDNet, because of its simple structure and robust performance. As we can see in Fig. 2, the quality of inputted images with adverse image conditions throughout the image-adaptive filter module can be significantly improved.

1) Image-Adaptive Filter Module: Generally speaking, the haze-remove or image-enhanced filter can improve the image quality, thus, we used five image filter operators including the haze-remove filter, exposure filter, gamma filter, contrast filter, and sharpen filter to improve the image quality. It's worth noting that the aforementioned filter operations are differentiable, which means that the parameters of these filter operations can be learned from the CNN module without manual setting. To decrease the model's complexity, we proposed a simple CNN module to generate the image-adaptive filter module. As shown in Fig. 2, we first resize the inputted image to 256 × 256 pixels size to improve computation efficiency, and then five convolution modules are used to extract features from the resized image (the stride size of each convolution module is 2). In the output layer of

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the image-adaptive filter module, we used a large convolution module (the kernel size is set as 8) to obtain five filter parameters. Regarding the bi-temporal change detection task, we utilized two independent image-adaptive filter modules to learn parameters from inputted bi-temporal images, and the whole process was end-to-end.

As we know, most of the open-sourced training samples were collected from high-quality remote sensing images, therefore, how to simulate haze-cover or lowlight image conditions is very important to train IA-CDNet. Wang et al. [12] proposed a haze image simulation and mosaic simulation method, which could simulate realistic haze cover images. Therefore, we also used the haze simulation method to simulate haze cover scenarios. To simulate low-light images, we used a simple power transformation to implement it. The simulation equation can be seen as

$$\text{low Light Img} = \text{Img}^{\gamma} \tag{1}$$

where the γ parameter is randomly generated from [1.2, 5.5]. In such a wide data range, it can simulate low-light image conditions at various levels. We have already generated a haze-cover image according to [12], and then we can use the dark channel prior (DCP) algorithm to recover the haze-free image. First, the haze image can use the following equation to describe:

$$\text{HazeImg}(x) = J(x)t(x) + A(1 - t(x)) \tag{2}$$

where $\text{HazeImg}(x)$ represents the haze cover image, and $J(x)$, $t(x)$, and A represents the haze-free image, transmittance image, and atmospheric light value, respectively. To obtain a haze-free image $J(x)$, we should know the value of parameter A and $t(x)$. According to [45], we can easily set the A value as the mean value of 1% top pixel value of $\text{HazeImg}(x)$. Then, the following issue is how to get the $t(x)$ image. According to the DCP algorithm, the $t(x)$ image can be obtained using the following equation:

$$t(x) = 1 - w \times \min_{y \in \Omega(x)} \left(\min_{c} \frac{\text{HazeImg}^{c}(x)}{A^{c}} \right). \tag{3}$$

In (3), the most important parameter is the w, which can control the haze removal degree. The c represents the image band of the inputted image, in most cases, it can be set to 3 in most cases. The $\Omega(x)$ represents the local region of the inputted image, generally, it can be set in the range from 15 to 45 pixels. According to (2) and (3), we can easily remove haze in $\text{HazeImg}(x)$ to get a clear image $J(x)$ based on the following equation:

$$J(x) = \frac{\text{HazeImg}(x) - A(1 - t(x))}{t(x)}. \tag{4}$$

For the other four filter operators. The exposure filter and gamma filter were defined as a simple multiplication and power transformation, respectively, which can enhance image quality in low-light conditions. To adjust the image contrast, we used the so-called contrast filter to enhance the inputted image. The image sharpness degree will also affect the model performance, thus the sharpen filter was chosen to improve the quality of the inputted image. The mathematical equation of the aforementioned five differentiable image filters can be seen in Table III.

2) Change Detection Module: In this manuscript, we aim to expand the performance of the change detection model in adverse image conditions. To verify that, we selected the SNUNet-CD model (see Fig. 3) as our change detection module to achieve change detection in bi-temporal images [24]. The main contribution of SNUNet-CD was that it took full advantage of the shallow-layer information, and this improvement indicated that it acquired excellent results in the edge area of the changed region. Another novelty point of SNUNet-CD was the ECAM Module, which further improved the training speed and inference accuracy. After the adverse bi-temporal images throughout the image-adaptive filter module, the filtered bi-temporal images will be input into the change detection module, then the final pixel-level change detection results can be obtained. It is worth noting that the change detection module in IA-CDNet could be replaced by other semantic segmentation models or change detection models easily, in other words, IA-CDNet was a general framework for providing a novel method to improve deep learning performance in adverse scenarios.

3) Model Training: To improve the generalization ability of IA-CDNet, we used two data augmentation methods to enhance the model generalization ability, including the traditional data augmentation methods, and the haze simulation or low-light simulation methods. Traditional data augmentation methods mainly contain image rotation, image flip, and adding gauss noise. Traditional data augmentation methods and haze simulation or low-light simulation methods were conducted with a 50% probability in the training stage. The detailed training procedure can be seen in Algorithm 1.

Algorithm 1 Image-Adaptive IA-CDNet Training Procedure

```
Initialize the model parameters of IA-CDNet using Kaming-initial method. The batch size and epoch parameter can be seen in Section 2.3.1.
for i in range (0, epoch) do
  for img in batch do
    if random.randint(0, 10) > 5 then
      Generate a haze-cover image by using the haze-simulated algorithm in [12].
      Generate a low-light image by using equation (1).
    else
      Using traditional data augmentation methods.
    end if
    Use Image-Adaptive Filter Module in IA-CDNet to obtain five differentiable image filter parameters.
    Output bi-temporal images by using five image filter operators with the automatically learned filter parameters.
    Use the SNUNet-CD to conduct change detection on the filtered images.
    The weighted cross-entropy and dice loss were used to optimize the model weight parameter.
  end for
end for
```

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TABLE III
EQUATION OF IMAGE FILTER OPERATORS

Filter type	The differentiable parameter	Equation
Haze-remove filter	W	$outImg = \frac{HazeImg - A(1-t)}{t}$, Where t can be obtained from equation (3). The w parameter in equation (3) is the differentiable parameter
Exposure filter	E	$outImg = 2^E \times Img$
Gamma filter	G	$outImg = Img^G$
Contrast filter	$alpha$	$outImg = alpha \times En(I) + (1 - alpha) \times I$, where, $I = 0.27 \times r + 0.67 \times g + 0.06 \times b$, $En(I) = \frac{1}{2}(1 - cos(\pi \times I))$
Sharpen filter	$numta$	$outImg = Img + numta \times (Img - GaussFilter(Img))$, where, GaussFilter represents the Gauss filter.

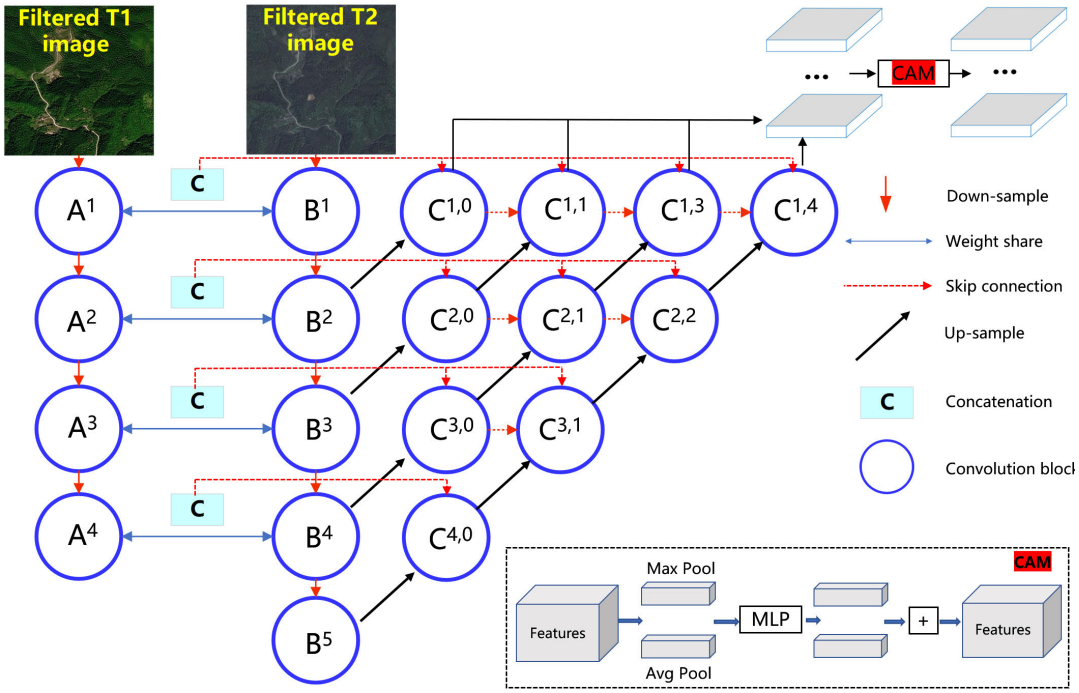


Fig. 3. Change detection module.

4) Accuracy Assessment: We used four indicators to evaluate the quantitative accuracy, and they were the recall, the precision, the $F1$-score, the overall accuracy, and the IOU [46]. The four indicators were widely used in change detection, and object extraction tasks on remote sensing imageries. The formula of the aforementioned indicators can be seen in the following equations:

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{5}$$

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{6}$$

$$F1\text{-score} = (2 \times \text{Recall} \times \text{Precision})/(\text{Recall} + \text{Precision}) \tag{7}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{8}$$

$$\text{IOU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}). \tag{9}$$

In the above equations, TP and TN mean the predicted results are the same as the GT, for example, the predicted result is a positive value, and the GT is also a positive value, which represents the TP indicator. FP means that the predicted result is a positive value, while the GT is a negative value. FN means the opposite result.

IV. RESULTS

A. Parameter Setting

The IA-CDNet was trained with Pytorch 2.2.0 with the Ubuntu 22.04 operation system. The init learning rate was set as 0.0001, and batch size and epoch were set as 8, and 60, respectively. The whole training process was on an RTX 3090 device with 24-GB RAM. The random seed was set as 3407 according to [47].

B. Loss Function

For change detection tasks in remote sensing images, the cross-entropy is the most widely used loss function to train a deep learning model. However, the class imbalance between

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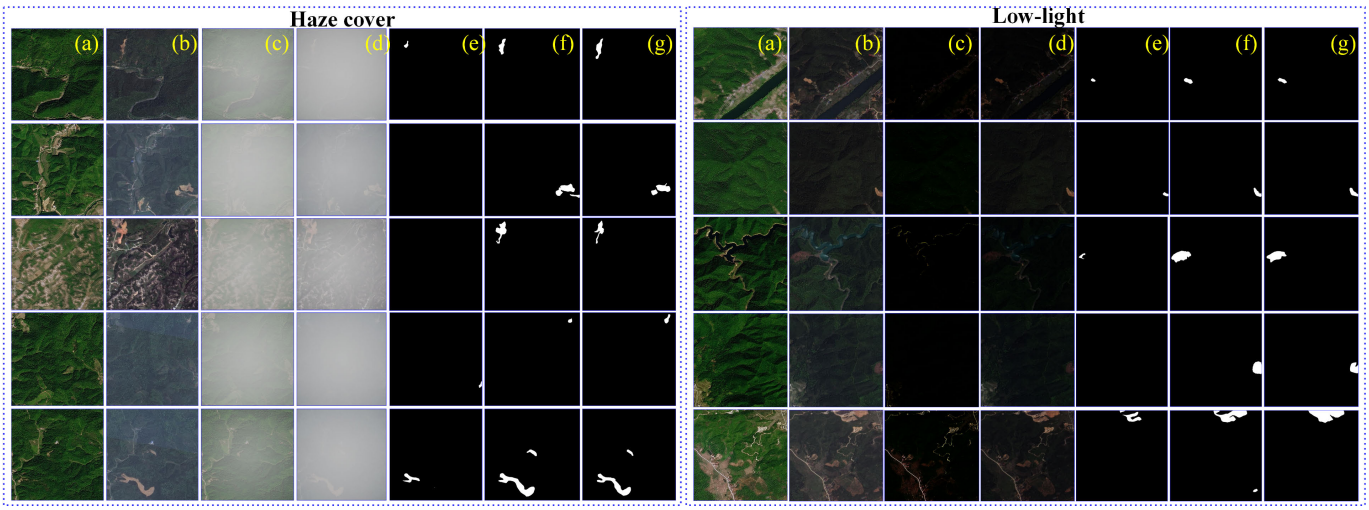


Fig. 4. Deforestation detection. (a) and (b) Original bi-temporal images. (c) and (d) Simulated bi-temporal images with haze-cover or low-light conditions. (e) Results by raw method. (f) Results by IA-CDNet. (g) GT.

positive pixels and negative pixels will significantly shrink the deep learning models' performance, thus we used the weighted cross-entropy [see (5)] and dice loss [see (6)] as our loss function to alleviate the class imbalance issue. Several research works also demonstrated that cross-entropy and dice loss achieved the best performance, especially for a pixel-level segmentation task [25]

$$\text{WCL} = -\sum_{c}^{C} w_c \times y_c \times \log(p_c), \quad \text{where}$$

$$w_c = \frac{\text{Num}(y_c = c)}{N} \tag{10}$$

$$\text{DL} = \frac{2 \times \sum_{i=1}^{N} p_i \times y_i}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} y_i^2} \tag{11}$$

$$\text{Loss} = \alpha \times \text{WCL} + (1 - \alpha) \times \text{DL}. \tag{12}$$

In the above equations, C represents the number of changed classes, P presents the predicted result by deep learning models, y represents the GT, and N means the number of pixels in a labeled image. The α represents the weight parameter, we set it as 0.5 to make these two loss functions keep the same weight.

C. Deforestation Detection Experiment

First, we defined the raw method as follows: the SNUNet-CD model trained with traditional data augmentation methods. To test the performance of IA-CDNet in adverse image conditions, we adopted the haze simulation method and low-light simulation method in Section III-B to simulate haze cover or low-light image conditions on the test dataset (a test image will be randomly transformed into haze cover or low-light conditions), as displayed in Fig. 4. On the whole, we can see that the IA-CDNet acquired a better visual result than the raw method. In addition, most of the deforestation in thick haze cover regions cannot be detected by the raw method. Furthermore, the deforestation detection results in low-light conditions also indicated that severe omission alarms occurred in the raw method. Deforestation detection experiments in adverse image conditions indicated that change detection models designed for high-quality image conditions may not work well in haze cover or low-light conditions. The carefully designed change detection mode, IA-CDNet can significantly improve the model performance in adverse image conditions. The quantitative accuracy assessment in Table IV also demonstrated the excellent performance of IA-CDNet.

D. Cropland Change Detection Experiment

We also selected the haze simulation and low-light simulation methods in Section III-B to simulate haze cover and low-light image conditions for cropland change detection, and the results were displayed in Fig. 5 (a test image will be randomly transformed into haze cover or low-light conditions). Regarding haze cover scenarios, we can see that the omission alarms occurred in the raw method, especially in the thick haze cover regions. The cropland change detection results in low-light image conditions showed that omission alarms and commission alarms occurred simultaneously. In the first and third rows of Fig. 4 (low-light condition), we can see that the degree of low-light cover was lower than in other images, and several objects with high digital number (DN) values were incorrectly detected as cropland change. The cropland change detection experiment also indicated that adverse image conditions would significantly decrease the model performance once again. The IA-CDNet displayed a much better visual result than the raw method, we can see that the changed regions with various shapes and sizes are accurately detected by IA-CDNet. The quantitative accuracy assessment in Table IV also proves the effectiveness of IA-CDNet.

E. LIM-CD Change Detection Experiment

In the first step, we also used the haze-simulated and low-light-simulated methods in Section III-B to randomly transform a test image into an adverse image condition. The change detection results on the LIM-CD dataset are presented

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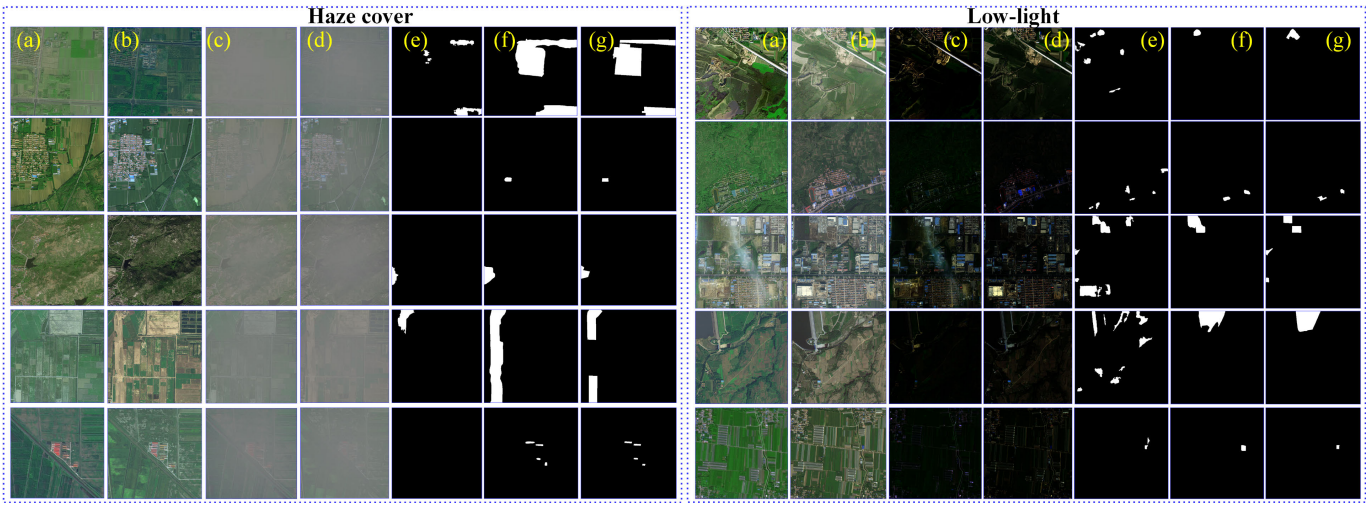


Fig. 5. Cropland change detection. (a) and (b) Original bi-temporal images. (c) and (d) Simulated bi-temporal images with haze-cover or low-light conditions. (e) Results by raw method. (f) Results by IA-CDNet. (g) GT.

TABLE IV
QUANTITATIVE ACCURACY ASSESSMENT ON THREE DATASETS

Datasets	Method	Recall	Precision	F1-score	OA	IOU
Deforestation detection	Raw method	0.6794	0.7891	0.4081	0.7321	0.3031
	Ours (IA-CDNet)	0.8293	0.9319	0.5838	0.9299	0.4721
Cropland change detection	Raw method	0.6853	0.7943	0.4271	0.7752	0.3078
	Ours (IA-CDNet)	0.8488	0.9481	0.6051	0.9561	0.4962
LIM-CD change detection	Raw method	0.6185	0.7285	0.3977	0.7001	0.3004
	Ours (IA-CDNet)	0.7631	0.8671	0.5118	0.8512	0.4181

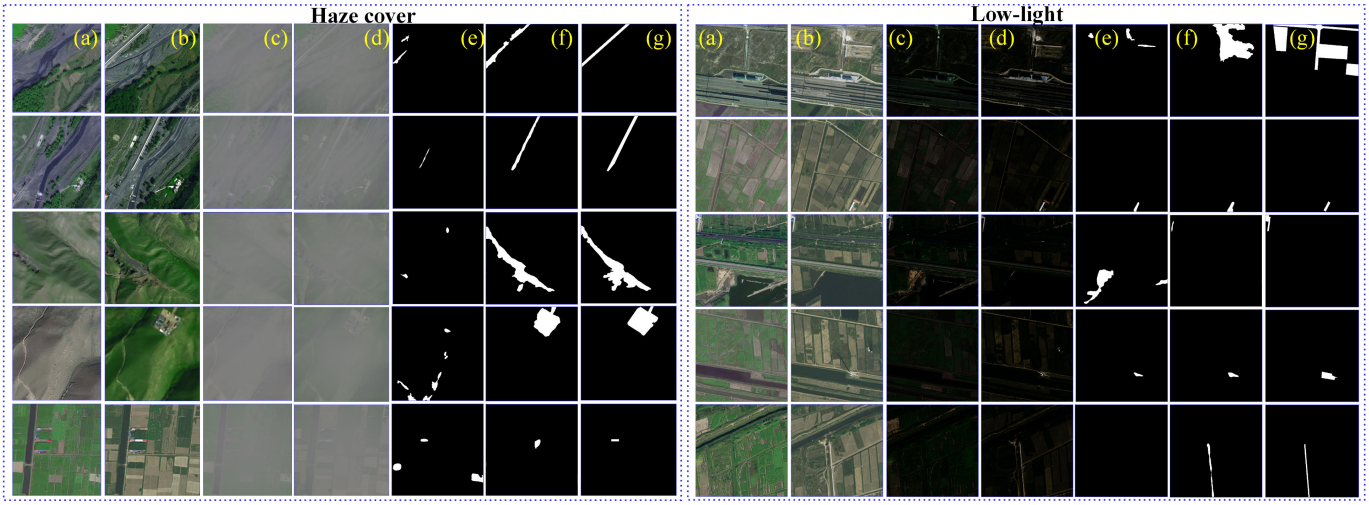


Fig. 6. Change detection on LIM-CD dataset. (a)–(d) Original and simulated bi-temporal images. (e)–(g) Results acquired by the raw method, MSBDN, the DehazeFormer, the haze simulation method, the DA-MSCDNet, the SDACD, BADNet, IA-CDNet, and GT.

in Fig. 6. On the whole, we can see that severe omission errors occurred in haze-cover scenarios, and several commission errors occurred in low-light scenarios with the raw method. In addition, we can clearly see that several change regions covered by thick haze failed to be detected by the raw method. For example, the newly constructed building in the fourth row (haze cover condition). In Fig. 6, only a few change regions with high DN values were successfully detected by the raw method, such as in the second and fourth rows of Fig. 6 (low-light condition). The quantitative accuracy assessment in Table IV also displayed that the IA-CDNet outperformed the raw method.

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TABLE V
COMPARED METHODS

Compared methods	Principle description
Raw method (SNUNet-CD) [24]	The SNUNet was selected as the baseline model to conduct change detection. To train this model, we only used general data augmentation methods, such as image rotation, flip, add gauss noise, etc.
MSBDN [16]	In this paper, the authors proposed a multi-scale boosted dehaze model, named MSBDN. We used the Haze simulation method in [12] to train MSBDN. In the inference stage, the change detection model will be applied to the haze-removed images. To compare fairly, we also used SNUNet-CD as the change detection model.
DehazeFormer [14]	This paper proposed a novel dehaze model, named DehazeFormer, which achieved state-of-the-art results on the RS-Haze dataset. It is worth noting that we finetune DehazeFormer with the haze cover dataset in [12]. The following steps are the same as MSBDN.
Haze and mosaic simulation method [12]	This paper proposed a haze and mosaic simulation data augmentation method to improve model performance in complicated regions, and it also used the SNUNet-CD as the baseline model.
BIT [48]	This paper proposed a transformer-style change detection model for bi-temporal images, named BIT. To train the BIT model in adverse image conditions, we used the haze cover or low-light image simulation method in Section 2.2.1 to train it.
ChangeFormer [49]	This paper introduced a transformer-based Siamese network for bi-temporal image change detection, named ChangeFormer. The main interesting point of ChangeFormer was that it used a Multi-Layer perception decoder to capture long-range semantic features. The training skill is the same as BIT.
SEIFNet [50]	This paper introduced an interesting spatiotemporal enhancement and interlevel fusion network, named SEIFNet, this model can leverage the multi-scale semantic features to improve change detection precision. Several experiments on open-sourced datasets validate its performance. We also used the same training skills as SDACD, or ChangeFormer to train SEIFNet in adverse image conditions.
DA-MSCDNet [19]	This paper proposed a domain adaption change detection model, namely DA-MSCDNet. This model can learn deep features from optical images and SAR images. To make DA-MSCDNet transfer into haze cover conditions, we used low-light or haze cover simulation introduced in Section 2.2.1 to generate adverse image pairs. In other words, the only change in DA-MSCDNet is the input image pairs. The following steps are kept the same as the original paper described.
SDACD [20]	his paper proposed a cross-domain change detection model, named SDACD. The SDACD model could make style transform between the bi-temporal remote sensing images, which significantly solved the spectral change caused by seasonal change or other factors. The following steps are the same as DA-MSCDNet.
BADNet [51]	This paper proposed an interesting object detection model for real-world haze scenes, named BADNet. Since the BADNet was designed for object detection tasks, thus we removed the object detection module in BADNet to make it adapt to change detection tasks on remote sensing images. We also selected the SNUNet as the change detection module to build the new BADNet, and then the following steps are the same as the original paper described.

V. DISCUSSION

In this section, we want to discuss the proposed image-adaptive method from the following aspects. First, how does it compare with other methods? Second, why does it can improve the model's performance in adverse image conditions? Third, how does the efficiency of it perform in the inference stage? Fourth, the importance of the proposed image filter modules. Fifth, how does IA-CDNet perform in real-world scenarios? Sixth, how does the image-adaptive module perform in other deep-learning models? Last but not least is the limitation of IA-CDNet.

A. Compared With Other Methods

What is the advantage of IA-CDNet when compared with the existing methods? We selected two excellent haze-remove models, three change detection models, and three domain adaption or image adaption models as the compared methods. The details of these methods can be seen in Table V.

The change detection results of three datasets are presented in Figs. 7–9. On the whole, we can see that the raw method acquired the worst result, and most of the change regions cannot be detected. This phenomenon meant that the traditional deep learning models have a limited ability to haze-cover adverse image conditions. The haze-remove method can be viewed as a useful image preprocessing method to improve the image quality before it is inputted into the deep learning models, we can see that both MSBDN [16] and DehazeFormer [14] can improve the deep learning model performance in haze-cover condition. However, they still cannot help the change detection models to detect some changes in thick haze-cover regions. The results of ChangeFormer, BIT, haze simulation method, and SEIFTNet indicated that the method can improve the model performance in haze-cover image conditions. Nevertheless, a few changed regions still cannot be detected accurately.

The domain adaption models indicated a better result than haze removal or mixed training method (haze simulation), we can explain it from two aspects: training skills and model structure. Indeed, we also used the mixed training method to train the domain adaption models, and this kind of training skill was already demonstrated effectively in the haze simulation method. Furthermore, the model structure of domain adaption methods was carefully designed, which can learn effective semantic features in haze-cover adverse image conditions. Therefore, the domain adaption method

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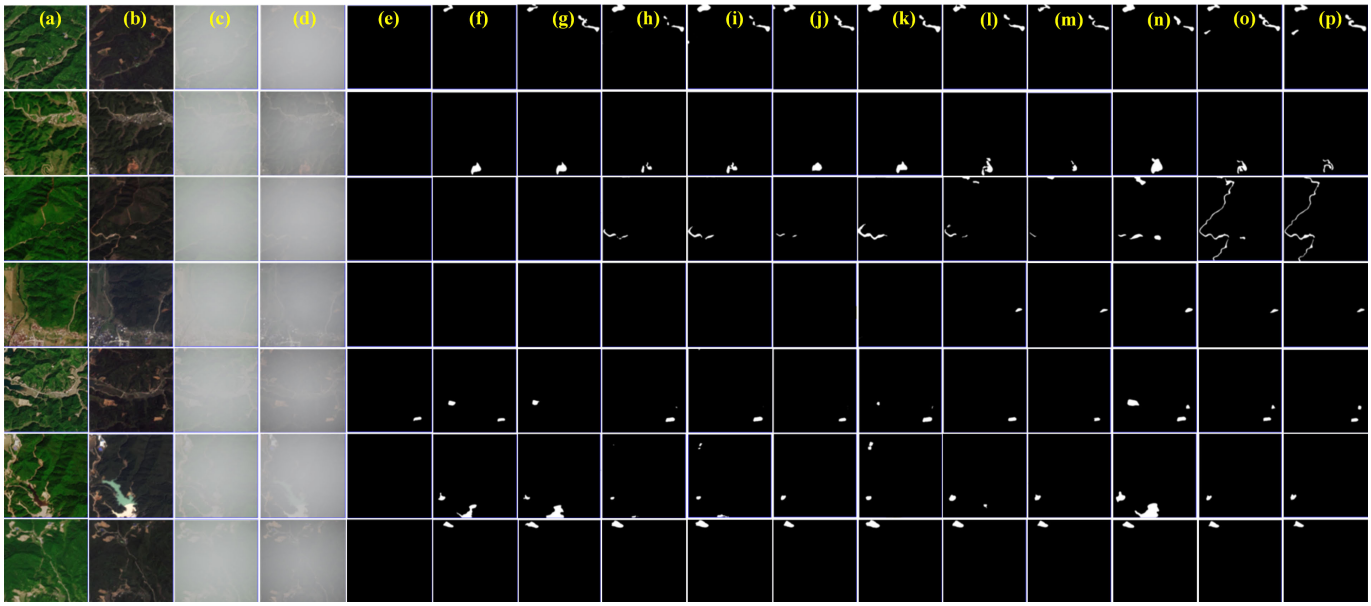


Fig. 7. Cropland change detection. (a)–(d) Original and simulated bi-temporal images. (e)–(p) Deforestation detection results by the raw method, MSBDN, the DehazeFormer, the ChangeFormer, the BIT, the haze simulation method, the SEIFNet, the DA-MSCDNet, the SDACD, BADNet, IA-CDNet, and GT.

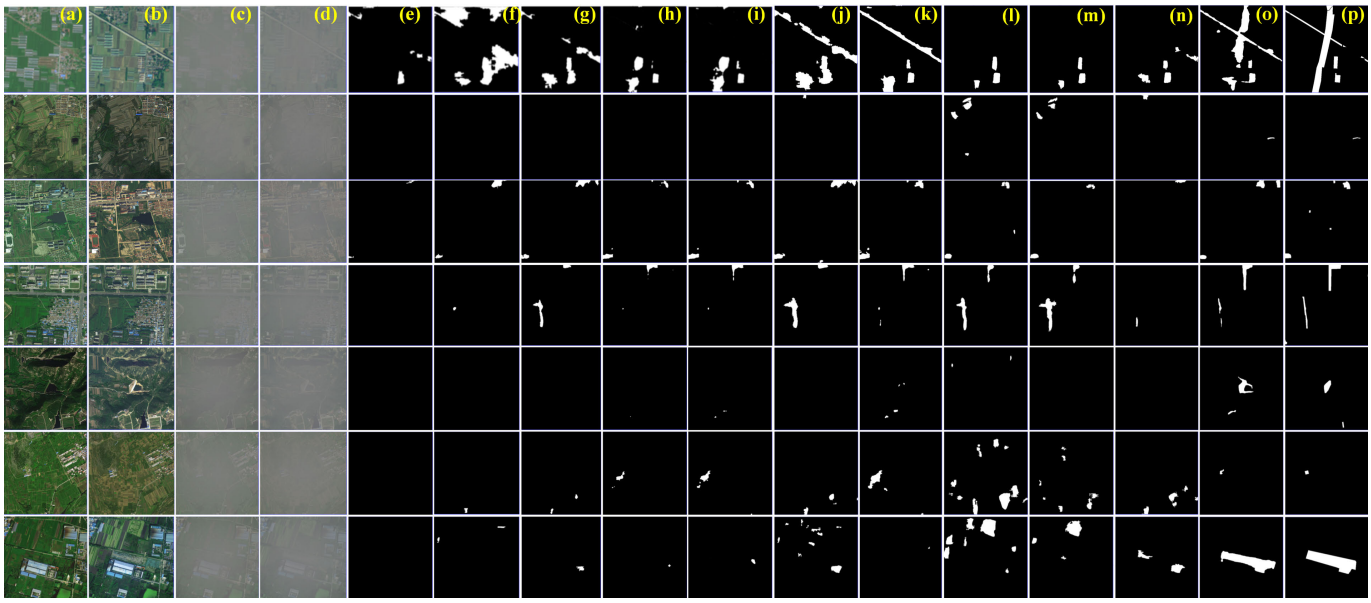


Fig. 8. LIM-CD change detection. (a)–(d) Original and simulated bi-temporal images. (e)–(p) Deforestation detection results by the raw method, MSBDN, the DehazeFormer, the ChangeFormer, the BIT, the haze simulation method, the SEIFNet, the DA-MSCDNet, the SDACD, BADNet, IA-CDNet, and GT.

also displayed an excellent performance in haze-cover image conditions. However, they still cannot detect slight changes successfully, such as for the narrow road constructions in the third row in Fig. 7. The change detection results obtained by the BADNet indicated that it achieved a slightly better performance than the two domain-adaptive models. However, compared with IA-CDNet, it still can be further improved. The performance of BADNet can be explained from the image enhancement aspect. Because the BADNet only has a dehaze module, thus the quality of the image throughout the dehaze module is still weaker than our image-adaptive module.

Our IA-CDNet indicated the best performance compared with the aforementioned models or methods, and the results showed that most of the changes are accurately detected by it. Moreover, compared with the domain adaption method, the model structure of IA-CDNet is simpler, because we only added an image-adaptive filter module in front of the change detection module. The quantitative accuracy assessment of various methods can be seen in Table VI, which also indicates the excellent performance of the proposed IA-CDNet.

B. Explanation of Image-Adaptive Method

Why does IA-CDNet can improve model performance under haze-cover or low-light adverse conditions? We also conducted an auxiliary experiment to describe. As displayed in Fig. 10, we can see that the inputted bi-temporal high-resolution remote sensing images were covered with haze, which makes the land object hard to identify. Nevertheless, after the bi-temporal images throughout the image-adaptive

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TABLE VI

ACCURACY ASSESSMENT OF COMPARED METHODS (TO OBTAIN FLOPS AND PARAMS, WE SET THE BATCH SIZE AND INPUT IMAGE SIZE AS 6, [3, 512, 512] PIXEL SIZE, RESPECTIVELY. THE RUNNING DEVICE WAS RTX 3090 AND AMD 5600X)

Datasets	Method	FLOPs(G)	Params(M)	Recall	Precision	F1-score	OA	IOU
Deforestation detection	Raw method	1315.99	12.03	0.6762	0.7883	0.3981	0.7240	0.3012
	MSBDN	2312.11	43.38	0.6814	0.8191	0.4291	0.7821	0.3351
	DehazeFormer	7915.02	37.43	0.7134	0.8274	0.4385	0.8283	0.3589
	ChangeFormer	279.69	29.75	0.7167	0.8298	0.4418	0.8314	0.3612
	BIT	1139.56	21.58	0.7204	0.8329	0.4561	0.8356	0.3645
	Haze simulation	1315.99	12.03	0.7284	0.8353	0.4732	0.8461	0.3712
	SEIFNet	200.88	27.91	0.7412	0.8401	0.5011	0.8529	0.4178
	DA-MSCDNet	2134.67	49.12	0.7456	0.8422	0.5194	0.8561	0.4261
	SDACD	2634.85	22.24	0.7732	0.8521	0.5329	0.8743	0.4398
	BADNet	1316.81	12.03	0.7889	0.8824	0.5571	0.8951	0.4472
	Ours (IA-CDNet)	1316.81	12.03	0.8293	0.9319	0.5749	0.9193	0.4619
Cropland change detection	Raw method	1315.99	12.03	0.6754	0.7802	0.4184	0.7649	0.3046
	MSBDN	2312.11	43.38	0.6874	0.7981	0.4284	0.7753	0.3311
	DehazeFormer	7915.02	37.43	0.7092	0.8071	0.4417	0.7921	0.3610
	ChangeFormer	279.69	29.75	0.7256	0.8173	0.4672	0.8156	0.3719
	BIT	1139.56	21.58	0.7123	0.8093	0.4561	0.8017	0.3678
	Haze simulation	1315.99	12.03	0.7293	0.8271	0.4869	0.8223	0.3821
	SEIFNet	200.88	27.91	0.7434	0.8376	0.5126	0.8345	0.4183
	DA-MSCDNet	2134.67	49.12	0.7591	0.8464	0.5288	0.8412	0.4338
	SDACD	2634.85	22.24	0.7782	0.8649	0.5551	0.8598	0.4519
	BADNet	1316.81	12.03	0.7923	0.8852	0.5631	0.8683	0.4681
	Ours (IA-CDNet)	1316.81	12.03	0.8329	0.9221	0.5902	0.9473	0.4876
LIM-CD change detection	Raw method	1315.99	12.03	0.6101	0.7194	0.3843	0.6941	0.2915
	MSBDN	2312.11	43.38	0.6291	0.7426	0.4172	0.7291	0.3298
	DehazeFormer	7915.02	37.43	0.6302	0.7562	0.4239	0.7431	0.3476
	ChangeFormer	279.69	29.75	0.6462	0.7712	0.4478	0.7543	0.3598
	BIT	1139.56	21.58	0.6394	0.7601	0.4311	0.7467	0.3517
	Haze simulation	1315.99	12.03	0.6521	0.7792	0.4582	0.7652	0.3651
	SEIFNet	200.88	27.91	0.6691	0.7843	0.4704	0.7712	0.2687
	DA-MSCDNet	2134.67	49.12	0.6728	0.7921	0.4752	0.7742	0.3701
	SDACD	2634.85	22.24	0.6982	0.8158	0.4821	0.7993	0.3811
	BADNet	1316.81	12.03	0.7134	0.8256	0.4898	0.8123	0.3912
	Ours (IA-CDNet)	1316.81	12.03	0.7542	0.8502	0.5065	0.8411	0.4101

filter module in IA-CDNet, we can see that the output bi-temporal images clearly display the morphology of land objects, thus the change detection model on the haze-cover region can detect changes more accurately. In Fig. 10, the low-light images condition also demonstrated that the image-adaptive filter module can easily improve the quality of

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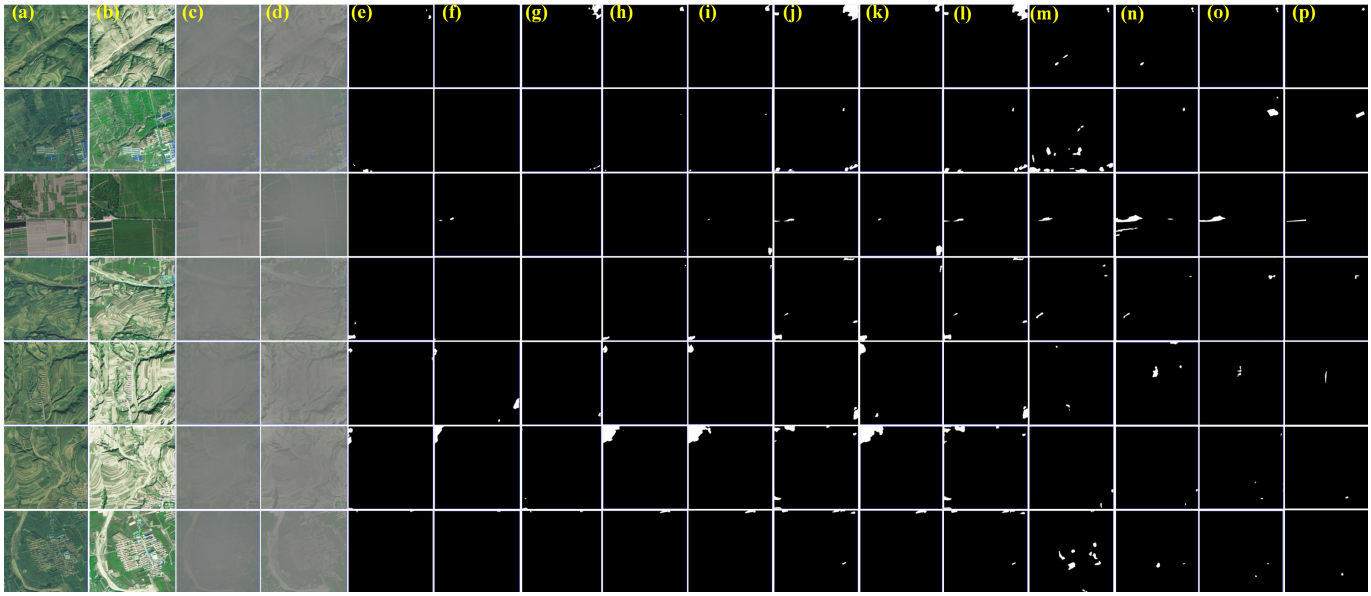


Fig. 9. LIM-CD change detection. (a)–(d) Original and simulated bi-temporal images. (e)–(p) Deforestation detection results by the raw method, MSBDN, the DehazeFormer, the ChangeFormer, the BIT, the haze simulation method, the SEIFNet, the DA-MSCDNet, the SDACD, BADNet, IA-CDNet, and GT.

TABLE VII

MODEL INFERENCE EFFICIENCY ASSESSMENT (THE BATCH SIZE AND INPUTTED IMAGE SIZE WERE 6, [3, 512, 512] PIXEL SIZE, RESPECTIVELY. THE RUNNING DEVICES INCLUDING THE GPU AND CPU DEVICES ARE RTX 3090, AND AMD 5600X, RESPECTIVELY)

Method	Parameters(M)	FLOPs(G)	Inference time on CPU device (s)	Inference time on GPU device (s)
Raw Method	12.03	1315.99	11.5670	0.6096
MSBDN	43.38	2312.11	21.7767	1.0912
DehazeFormer	37.43	7915.02	25.1243	2.6781
ChangeFormer	29.75	279.69	7.6369	0.3975
BIT	21.58	1139.56	6.7324	0.2876
Haze simulation	12.03	1315.99	11.5670	0.6096
SEIFNet	27.91	200.88	1.9763	0.2158
DA-MSCDNet	49.12	2134.67	17.1923	0.8032
SDACD	22.24	2634.85	18.43	0.9012
BADNet	12.03	1316.81	15.9812	0.7912
LA-CDNet	12.60	1316.92	16.7604	0.8009

inputted bi-temporal images. As a whole, the IA-CDNet can significantly improve its performance in the haze-cover or low-light image condition, mainly because the image-adaptive filter module can automatically restore high-quality images by learning accurate image filter parameters.

C. Efficiency Test

As we know, model inference efficiency is an important indicator to evaluate the performance of a change detection model. In IA-CDNet, the image-adaptive filter module is an auxiliary module, thus, how it affects the model inference efficiency should be carefully investigated. The model parameter and inference time are presented in Table VII, we can see that the parameter and FLOPs indicator showed that the image adaptive-filter module was lightweight. In addition, the difference in inference time on the GPU device between the raw method and IA-CDNet was also very small. However, the inference time of two models in the CPU device seemed a little large, maybe the image filter operators should be carefully optimized on the CPU device. As we know, most of the advanced deep-learning models will be deployed on GPU devices, especially for the change detection models for high-resolution remote sensing images. In Table VII, we can see that the IA-CDNet was a GPU-friendly model, thus considering the tradeoff from the accuracy and computation resource occupied, IA-CDNet was an excellent change detection model to be used on adverse image conditions.

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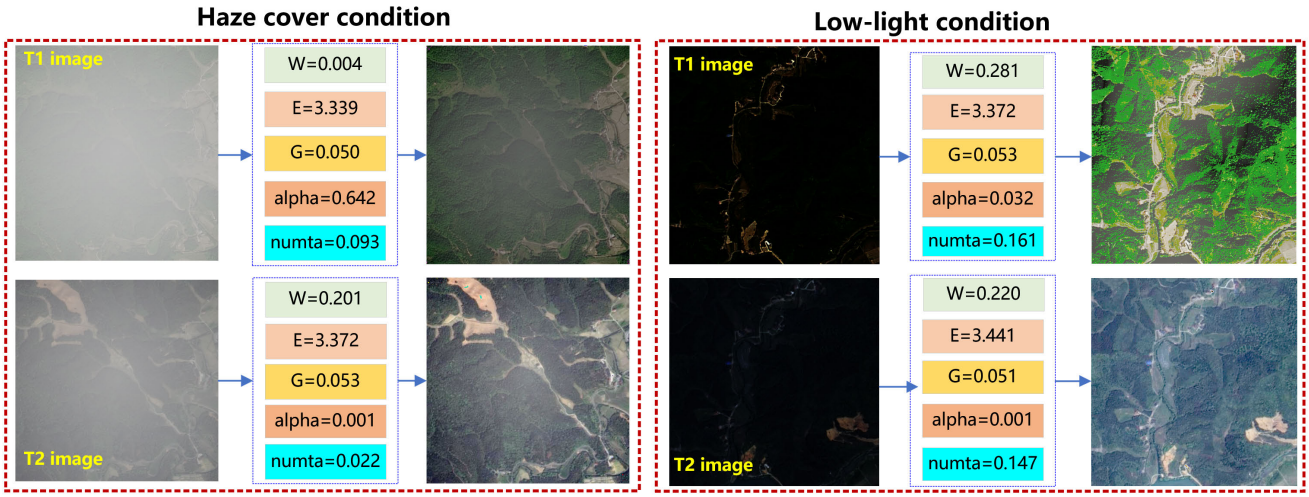


Fig. 10. Image filtered through the image-adaptive filter module (haze cover image condition and low-light image condition are displayed in the left column and right column, respectively).

TABLE VIII
ABLATION STUDY

Method	haze-remove filter	Exposure filter	Gamma filter	Contrast filter	Sharpen filter	F1-score
Raw method (mixed training)						0.5027
IA-CDNet	✓					0.5391
IA-CDNet	✓	✓				0.5421
IA-CDNet	✓	✓	✓			0.5543
IA-CDNet	✓	✓	✓	✓		0.5695
IA-CDNet	✓	✓	✓	✓	✓	0.5838

D. Ablation Study

In this section, we want to discuss how the five image filters affect the accuracy of change detection. We have conducted an ablation study to describe it. The deforestation detection experiment was selected to validate, and the adverse image conditions such as haze cover and low light were simulated using the method proposed in Section III-B. We selected the SNUNet-CD as the baseline model, and it is worth noting that the SNUNet-CD model was trained with mixed training samples skills, in other words, the original high-quality samples will be mixed with the simulated haze-cover and low-light samples. Then, the five image filter operators were added iteratively to test the performance of each image filter operator. The ablation study results can be seen in Table VIII. We can see that all five image filter operators contributed to improving the model performance, interestingly, the haze-remove filter and Sharpen filter occupy the most important position. This phenomenon provided some inspiration for us that designing an advanced and effective image filter operator was important to enhance the model's ability.

E. Experiments in Real-World Scenarios

How does IA-CDNet perform in real-world scenarios? We also selected real-world scenarios including haze-cover and low-light image conditions to describe. The detailed image information of experimental images is presented in Table II. It's worth noting that we used a moving window to read sliced images from large-scale images to avoid the memory-limited issue.

1) Performance in Real-World Haze-Cover Scenarios: We selected deforestation detection as the experiment scenario. The bi-temporal high-resolution remote sensing images used to detect deforestation are displayed in Fig. 11. We can see that the former time-phase image was covered with haze, and the low-left region of the image was covered with thick haze, which made the land objects even harder to identify. The GT map was generated by visual interpretation from the reference image and the latter time-phase image. The difference of image acquired time between the former time-phase image and the reference images was only one day, thus, the reference image can be defined as the high-quality former time-phase image.

As a whole, the deforestation detection results with IA-CDNet indicated a better visual effect. The raw method displayed a few false-detected results in the low-left region, this phenomenon was mainly because of its poor generalization ability in adverse scenarios. To display the detected result clearly, we selected four subregions with various haze cover degrees to describe. The deforestation detection result of four subregions can be seen in Fig. 12.

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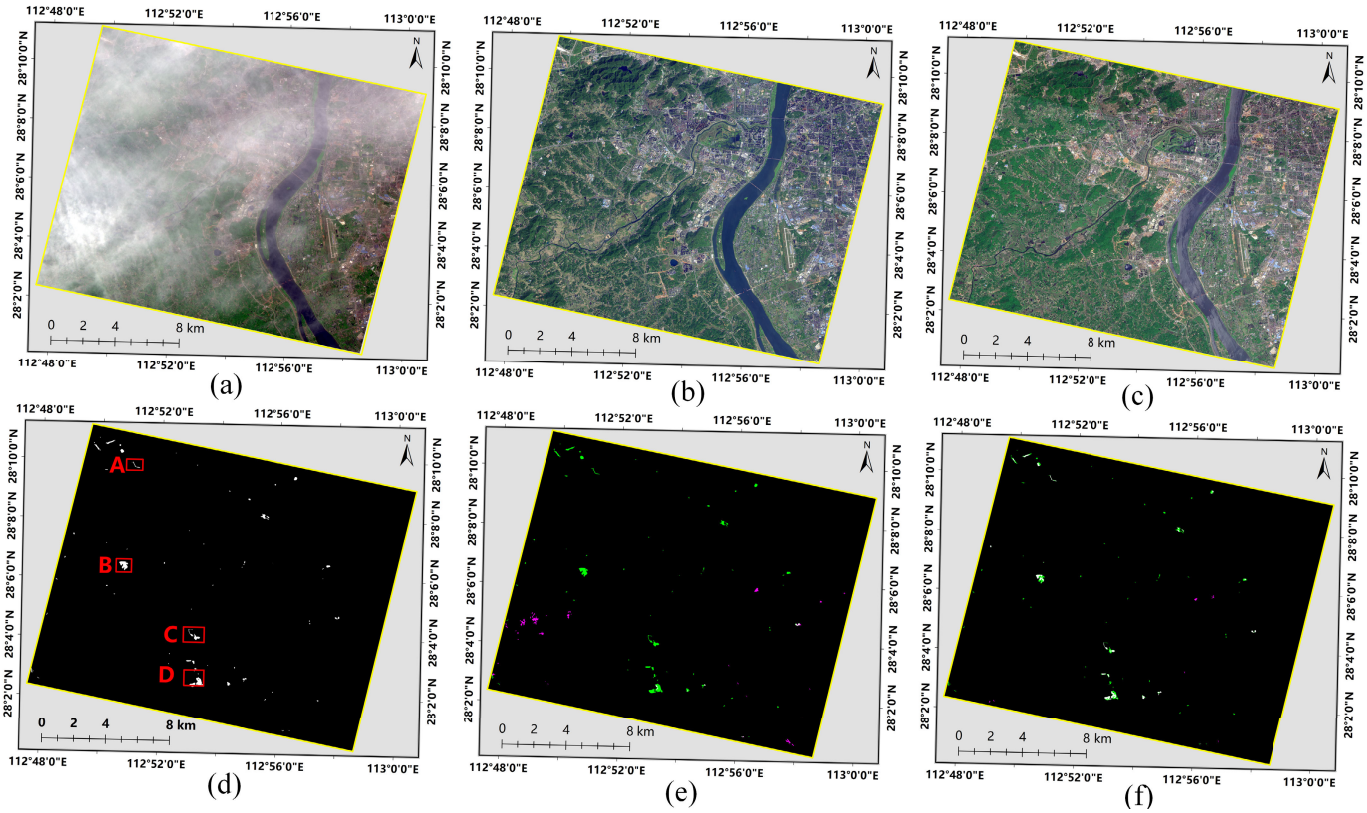


Fig. 11. Deforestation detection in real-world scenarios. (a) and (b) Bi-temporal images (the acquired time of them is 2023-4-15, and 2023-10-17, respectively). (c) Reference image (acquired on 2023-4-16). (d) GT. (e) and (f) Deforestation detection with the raw method, and IA-CDNet, respectively. (The green color and magenta color represent the omission and commission errors, respectively).

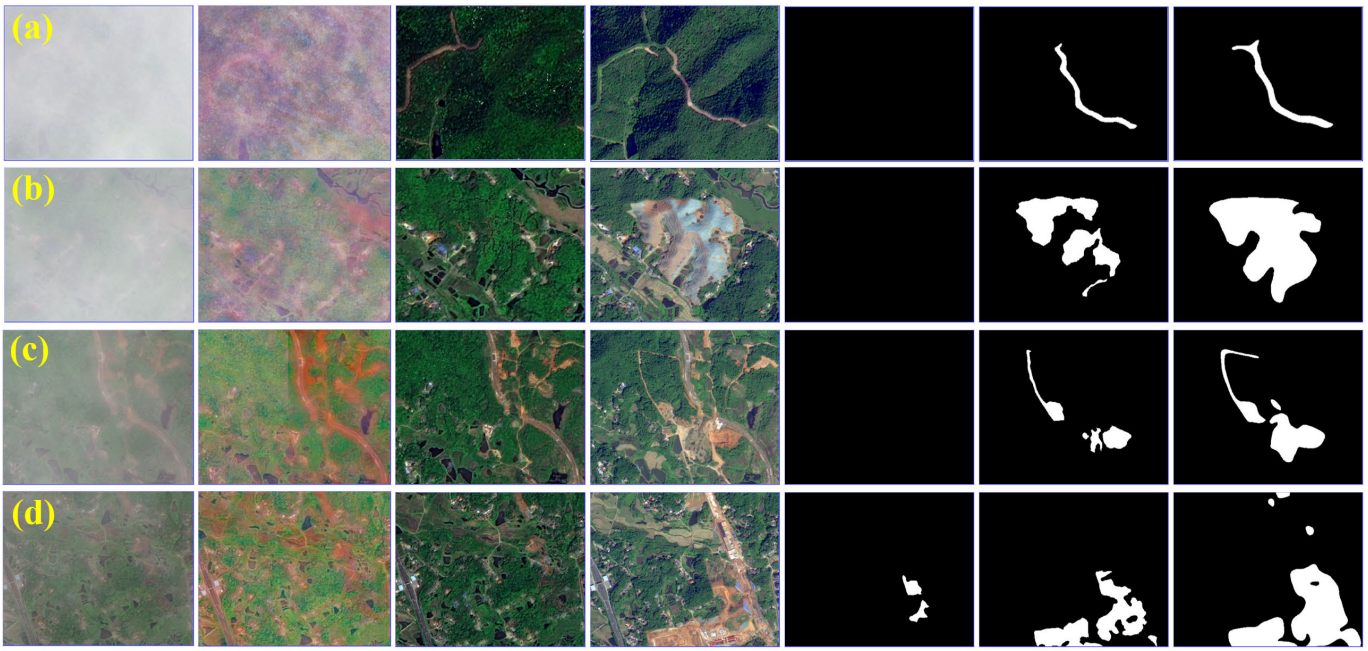


Fig. 12. (a)–(d) Deforestation detection in subregion A–D. (From left to right are the original former time-phase image, the filtered former time-phase image through the image-adaptive filter module, the reference image, the latter time-phase image, deforestation detection results by the raw method, deforestation detection results by IA-CDNet, and GT image, respectively).

In Fig. 12, we can see that the haze cover degree from A to D became less and less, and the deforestation detection result became better. However, in subregions A to C, the raw method cannot detect any changes, while IA-CDNet detected changes accurately, even in region A with thick haze cover scenarios. In the second row of Fig. 12, we can see that the image quality of the output image from former time-phase images was significantly improved, in subregion A, the raw image quality of the former time-phase image was covered with thick haze, which made the road construction objects hard to interpret, nevertheless, the image quality was improved obviously by using the image-adaptive filter module in IA-CDNet. Generally speaking, the IA-CDNet can significantly improve the image quality in the haze-cover regions and

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TABLE IX
QUANTITATIVE ACCURACY ASSESSMENT ON THREE DATASETS

Tasks	Method	Recall	Precision	F1-score	OA	IOU
Deforestation detection	Raw method	0.4012	0.3245	0.2921	0.9771	0.1952
	Ours (IA-CDNet)	0.7213	0.6221	0.5924	0.9889	0.4631
Cropland change detection	Raw method	0.3941	0.3051	0.2343	0.9601	0.1652
	Ours (IA-CDNet)	0.7061	0.6012	0.5178	0.9771	0.3619

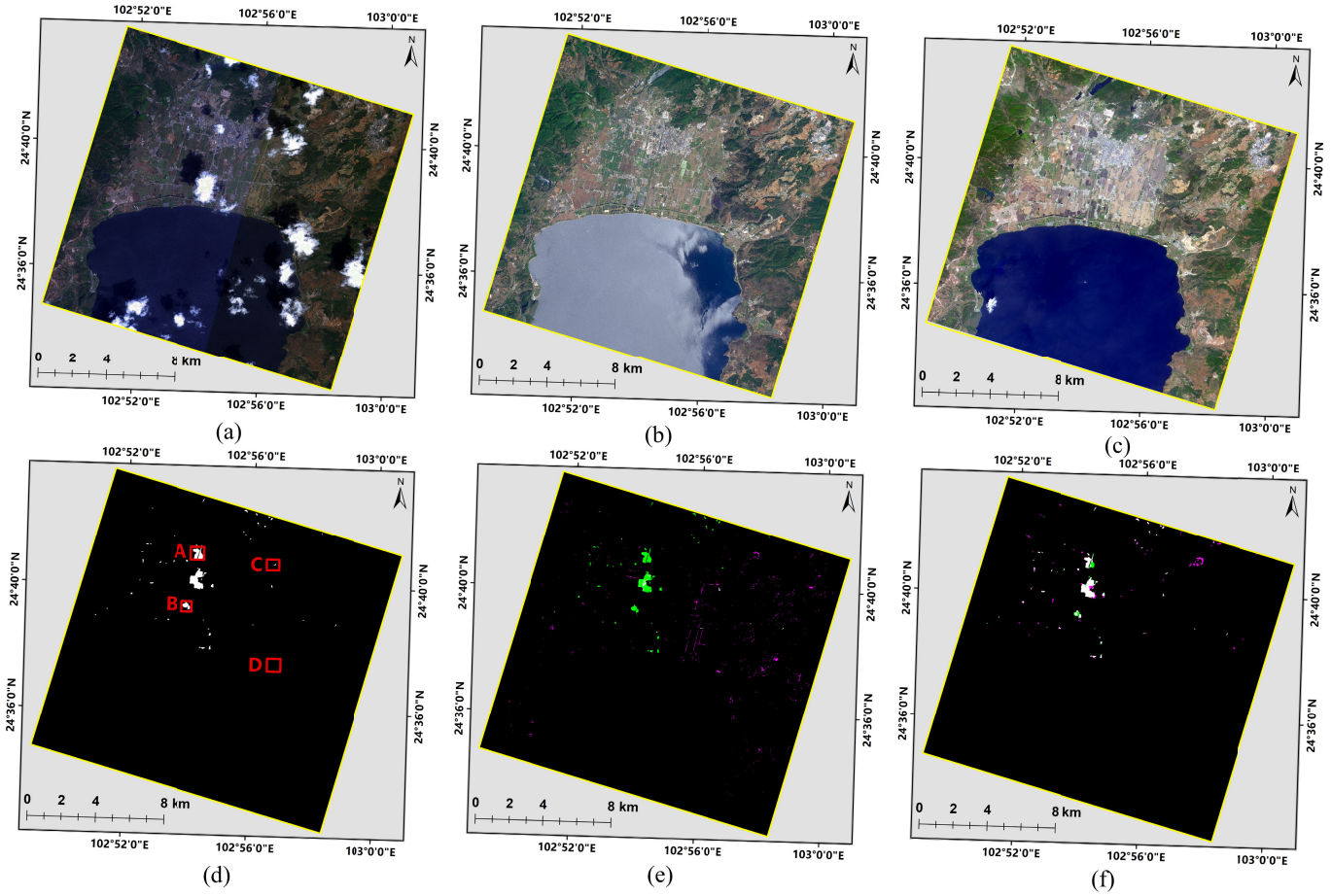


Fig. 13. Cropland change detection in real-world scenarios. (a) and (b) Bi-temporal images (the acquired time of them is April 2, 2019 and April 6, 2023, respectively). (c) Reference image (acquired on April 27, 2019). (d) GT. (e) and (f) Cropland change detection with the raw method, and IA-CDNet, respectively (the green color and magenta color represent the omission and commission errors, respectively).

improve the change detection model performance in such adverse scenarios. The quantitative accuracy assessment in Table IX also proved that the IA-CDNet had a better performance than the traditional change detection model. In Table IX, we can see that the $F1$-score indicator of IA-CDNet indicated a much better result than the raw method. The OA indicator in both IA-CDNet and the raw method displayed a high value because the number of background pixels was much higher than the positive pixels [52].

2) Performance in Real-World Low-Light Scenarios: We selected the cropland change detection as the experiment scenario. The cropland change detection in real-world scenarios is presented in Fig. 13. We can see that the former time-phase image was covered with cloud and cloud shadow, and the region covered with cloud shadow regions displayed a low-light image condition, which brought a challenge for change detection in these regions. Also, we selected a high-quality image as the reference image of the former time-phase image. The time difference of image acquired time between the former time-phase image and the reference image is very short, thus the reference image can be defined as the high-quality former time-phase image. The GT map was generated from the latter time-phase image and the reference image using visual interpretation.

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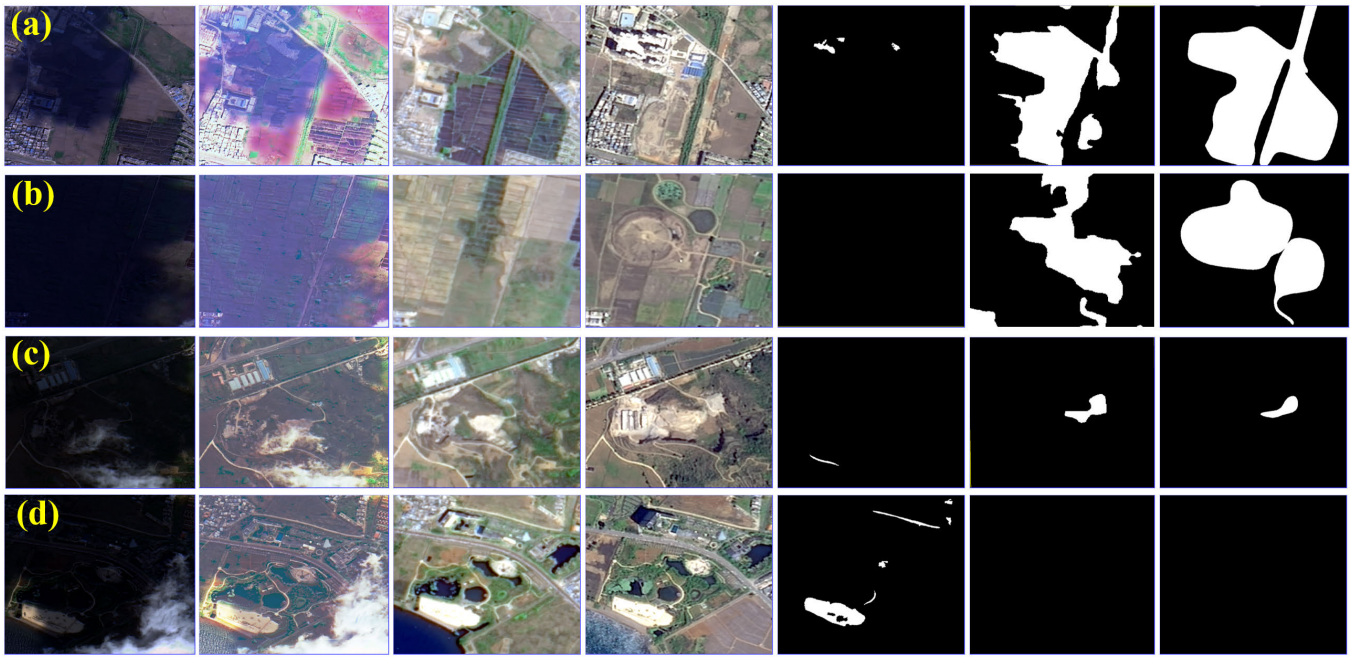


Fig. 14. Cropland change detection in subregion A–D (from left to right are the original former time-phase image, the filtered former time-phase image through the image-adaptive filter module, the referred image, the latter time-phase image, cropland change detection results by the raw method, cropland change detection results by the IA-CDNet, and GT image, respectively).

TABLE X
PERFORMANCE OF THE IMAGE-ADAPTIVE MODULE IN YARIOUS CHANGE DETECTION MODELS

Method	Recall	Precision	F1-score	OA	IOU
DTCDSCN (Raw method)	0.6682	0.7701	0.3921	0.7134	0.3001
DTCDSCN (+ image-adaptive module)	0.8378	0.9459	0.5743	0.9154	0.4611
ChangeFormer (Raw method)	0.6891	0.8012	0.4123	0.7481	0.3275
ChangeFormer (+ image-adaptive module)	0.8467	0.9561	0.6081	0.9487	0.4801
ConvTransNet (Raw method)	0.6912	0.8094	0.4207	0.7509	0.3287
ConvTransNet (+ image-adaptive module)	0.8498	0.9589	0.6103	0.9512	0.4817

On the whole, we can see that the cropland change detection result with IA-CDNet indicated a better visual result than the raw method, especially in the shadow region. To describe the difference of detected change clearly, we also selected four subregions to compare, and the result can be seen in Fig. 14. Obviously, the detected changes by using the raw method displayed a much higher omission error than IA-CDNet, because most of the cropland changes were not detected successfully in these shadow cover regions with the raw method. Regarding the low-light condition in the shadow cover region, the IA-CDNet can easily restore high-quality images before it was inputted into the change detection module. As displayed in the second row of Fig. 14, we can see that the image quality has significantly improved by using the image-adaptive filter module in IA-CDNet. Therefore, the accuracy of cropland change detection in the low-light condition was easily improved. The quantitative accuracy assessment of cropland change detection can be seen in Table IX, and it showed that IA-CDNet had a robust performance in low-light adverse conditions.

F. Performance of Image-Adaptive Module in Other Deep Learning Models

As previously described, the image-adaptive module can be integrated into any deep learning model, thus, we selected several advanced change detection models to validate this module's performance. The DTCDSCN (CNN-style) [50], the ChangeFormer (transform-style) [53], and the ConvTransNet (CNN-transformer mixed style) [7] of deep learning models were selected to validate. In the first step, we also defined the raw method as the change detection models trained with traditional data augmentation methods. Then, we added the image-adaptive module in front of the aforementioned change detection models to build modified models. Finally, we selected deforestation detection as the experiment dataset to validate the performance of the image-adaptive module (the test dataset was the same as it was in Section III). The performance of the image-adaptive module can be seen in Table X. On the one hand, we can see that the image-adaptive module can significantly improve the model performance in

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haze-cover and low-light conditions. On the other hand, the image-adaptive module also performed well in various model structures, thus, it can be defined as a general module to improve change detection models' performance in adverse image conditions.

G. Limitation

Although IA-CDNet performed well in haze-cover or low-light conditions, its performance in thick cloud-cover regions is still limited. As we know, the shape and semantic information of land objects are missing and this issue is almost no solution for single time-phase images in thick cloud-cover regions at present. In our future work, we may use multiple remote sensing images to improve the change detection accuracy in thick cloud cover regions [54].

VI. Conclusion

In most traditional change detection scenarios on bi-temporal remote sensing images, high-quality images without haze-cover scenarios are the first choice for researchers. However, high-quality images are not always provided for all scenarios, especially for emergency disaster scenarios, thus developing an advanced change detection model for adverse image conditions has a significant value. The main contribution of this manuscript is that we proposed a novel change detection model, named IA-CDNet, which is an advanced model designed for adverse remote image conditions, such as haze cover or low-light conditions, it is noted that the two scenarios are the most frequent degradation condition on optical remote sensing images.

The IA-CDNet is an image-adaptive model and its advantage is the model explainability. We designed five differentiable image filter operators to improve the inputted image quality, and the image filter parameter can be learned by using a small CNN module, which is not only compact but also readily integrated into any existing change detection models to expand their ability in adverse scenarios. Three open-sourced datasets and two large-scale real-world scenarios experiments demonstrated that IA-CDNet can significantly improve the change detection model performance. Compared with the state-of-the-art model, the IA-CDNet also indicated a superior performance. The IA-CDNet is easy to implement, and we hope this work can provide some new insights for researchers to conduct change detection in adverse scenarios.

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